

13	D
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14	B
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15	B
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Question	marking guidance	Additional comments/guidelines	mark
04.1	Uses $\omega = 2\pi f$ with $f = \frac{1}{T}$ or uses $\omega = \frac{2\pi}{T}$ directly ✓ $2.66 \times 10^{-6} \text{ (rad s}^{-1}\text{)} \checkmark$	$T = 2.36 \times 10^6 \text{ s}$	2
04.2	Uses $g = \frac{Gm}{r^2}$ for Earth ✓ $2.0 \times 10^{-3} \text{ (N kg}^{-1}\text{)} \checkmark$	Condone POT in r	2
04.3	Uses $a = \omega^2 r$ ✓ $3.3 \times 10^{-3} \text{ (m s}^{-2}\text{)} \checkmark$	Accept $3.2 \times 10^{-3} \text{ (m s}^{-2}\text{)}$. Allow 3.3×10^{-6} if failure to convert km to m already penalised in 04.2.	2
04.4	Two from: ✓ ✓ Idea that satellite's orbital period is lower than expected for this orbit or should be greater (at that distance/in absence of Moon) Idea that orbit requires greater centripetal force than that provided by Earth alone Idea that field strengths/forces of Earth and Moon will add (to provide required centripetal acceleration) Idea that effective mass (in Newton's gravitational law) or gravitational/centripetal force is larger	Orbital speed may be referenced instead of orbital period. Effective mass is combined mass of Earth and Moon	2
04.5	Line passing at right angles to all equipotential lines ✓	Condone 1 out of 5 crossings not at a right angle. Ignore any arrows.	1
Total			9

Question	Marking guidance	Additional comments/guidelines	Mark
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12	C
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13	D
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20	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	The work done per kilogram ✓ in bringing a mass from infinity to the surface (of Io) ✓ is $-3.27 \times 10^6 \text{ J kg}^{-1}$.	the idea of per kilogram can be seen in either MP1 or MP2 per unit mass is only sufficient for MP1 if J kg^{-1} is seen as a unit Number is required for two marks, but partial credit can be given for either line without it.	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	use of $V = -\frac{GM}{r}$ to give $1.82 \times 10^6 \text{ (m)}$ ✓	at least 3sf	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	use of potential energy = mV OR use of $\frac{1}{2}mv^2$ ✓ adds initial gpe and initial ke ✓ to give $-8.9(4) \times 10^6 \text{ (J kg}^{-1}\text{)}$ ✓	algebra must be seen for MP1 condone sign for mp1 signs must be correct for mp2 condone no algebra for MP2 at least 2 sf	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	use of $E = -\frac{GMm}{r}$ ✓ subtracts radius of moon to give $h = 1.8 \times 10^5$ (m) ✓	$h = 1.9 \times 10^5$ (m) if show-that data used	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	the height would be the same ✓ idea that mass of the projectile cancels out in the conservation of energy equation <i>or</i> both KE and GPE are directly proportional to mass ✓	For MP2, accept shown algebraically, but do not credit algebra that includes $GPE = mgh$	2	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	B
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Evidence of use of $g = \frac{GM}{r^2}$ ✓ To give $1.45 \text{ (m s}^{-2}\text{)}$ ✓	Condone POT error in substitution	2	2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Use of $\omega = \sqrt{\frac{GM}{r^3}}$ and $T = \frac{2\pi}{\omega}$ ✓ To give $7.1 \times 10^3 \text{ (s)}$ ✓	Alternative: use of $a = \omega^2 r$ and $T = \frac{2\pi}{\omega}$ ✓ Allow ecf for their a in 5.1 to calculate T ✓ In MP1 condone one substitution error for M or r.	2	AO1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Use of GPE = mV (to calculate V) ✓ Use of gravitational potential $V = \frac{GM}{r}$ ✓ To give $4.7 \times 10^4 \text{ (m)}$ ✓	Expect to see $r = 1.78 \times 10^6 \text{ m}$ Condone $4.3 \times 10^4 \text{ (m)}$ if rounded value of r used	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Calculation of KE in orbit OR calculation of GPE on surface of Moon ✓ Evidence of (KE + GPE) in orbit – (GPE on surface of moon) ✓ To give $1.83 \times 10^8 \text{ J}$ ✓	Expect to see $1.74 (\times 10^8 \text{ J})$ Expect to see $(-)3.56 (\times 10^8 \text{ J})$	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	Relates gravitational field strength to centripetal acceleration. ✓ Identifies that increase (local) density increases gravitational field strength (and therefore increases centripetal acceleration) ✓	MP2 could be argued from reduction in distance to centre of mass of Moon If no other mark awarded allow an application of Newton's 2nd Law of motion.	2	AO3
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
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19	D	$r \propto \frac{1}{v^2}$	AO1
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21	C	4.9 N kg^{-1}	AO2
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Question	Marking guidance	Mark	Comments
02.1	Starts with $\frac{GMm}{r^2} = \frac{mv^2}{r}$ ✓ Substitution of $KE = \frac{1}{2}mv^2$ leading to $KE = \frac{GMm}{2r}$ ✓ Adds $GPE = -\frac{GMm}{r}$ ✓	3	Mp1 must be a combination of gravitational force and centripetal force

Question	Marking guidance	Mark	Comments
02.2	Any two: Idea that satellite requires energy to move away from the Moon AND potential energy is zero at $r = \text{infinity}$ ✓ It is a bound system because of Moon's gravitational attraction OR magnitude of KE is less than magnitude of negative GPE wtte ✓	2	MP1 explains why GPE is negative MP2 explains why overall energy is negative

Question	Marking guidance	Mark	Comments
02.3	some form of $v = \sqrt{\frac{GM}{r}}$ ✓ correct substitution seen including correct mass ✓ $= 1.58 \times 10^3 \text{ (m s}^{-1}\text{)}$ ✓	3	MP1 can be squared and can be algebraic or with incorrect mass substituted Substituted mass must be Moon's ($7.3 \times 10^{22} \text{ kg}$) in MP2

Question	Marking guidance	Mark	Comments
02.4	Thrust/force is opposite to motion so work is done reducing total/gravitational potential energy ✓ link to r -dependence of total energy equation ✓	2	Treat any references to kinetic energy as neutral

Question	Marking guidance	Mark	Comments
02.5	Use of <i>difference in total energy</i> = $-\frac{GMm}{2r_2} - -\frac{GMm}{2r_1}$ ✓ = 6.7×10^7 (J) ✓	2	Subtraction can have the energy terms in either order If GPE is used in MP1, relationship to total energy must be clear (calculator value 6.748579×10^7 J)

Total		12
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Question	Marking guidance	Mark	Comments
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12	A	3.7 N kg^{-1}

Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	$F = G \frac{m_1 m_2}{r^2}$ seen and used To give $F = 3.88 \times 10^{12} \text{ N}$ ✓	Condone use of r = distance to Sun + radius of Sun to give $3.82 \times 10^{12} \text{ N}$ Calculator gives 3.879758×10^{12}	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	(For circular motion, $F = \frac{mv^2}{r}$) Determines $\frac{mv^2}{r}$ for comet ✓ indicates that this is not equal to force from 07.1 ✓	Expect to see $7.62(3) \times 10^{12}$ Allow similar calculations to show that one of r , v or d is not consistent with circular motion. Attempt to show m not correct should be given no credit.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Using 1 year for T and 1 Earth orbital distance for R $k = 1$ ✓ So $T^2 = k(18)^3$ ✓ $T = 76$ years ✓	Accept alternative values for k based on Earth orbit Accept missing k in MP2 Alternative using simultaneous equations MP1 equation applied to earth and comet e.g. $T_E^2 = kR_E^3$ $T_C^2 = kR_C^3$ OR $T_C^2 = k(18R_E)^3$ MP2 combines to remove k MP3 correct answer	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Calculates KE using $\frac{1}{2}mv^2$ OR Calculates GPE using $-G \frac{m_1 m_2}{r}$ OR Attempts to add KE and GPE numerically or algebraically ✓ To give $-6.0 \times 10^{21} \text{ J}$ ✓	Expect to see $3.338 \times 10^{23} \text{ J}$ Expect to see $-3.398 \times 10^{23} \text{ J}$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.5	Suggestion is correct Maximum/at infinity GPE is zero ✓ Idea that KE is greater than the energy needed to reach infinity/escape ✓	For MP2 allow idea that KE is greater than the magnitude of GPE	2	AO3
Total			10	

Question	Key	Answer	AO
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10	C	9	AO2
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11	D	$\frac{3g}{2\pi dG}$	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Idea that zero of E_p is at infinity ✓ idea that E_p increases as d increases/moving towards infinity OR Work is done as d increases/for object to reach infinity ✓	For MP2 allow idea that the field is attractive only if MP1 is given Allow reverse argument in MP2. Allow references to potential for E_p	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	At maximum, (force of) attraction from Moon = (force of) attraction from Earth OR (resultant) gravitational field strength is zero ✓ E_p increases going away from Moon when attracted to Moon OR E_p decreases towards Earth when attracted to Earth ✓	If no other mark awarded allow MAX 1 for idea that the gravitational field strength of the Earth is greater than the gravitational field strength of the Moon when $d > 4 \times 10^7$ (m)	2	AO1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Evidence of tangent drawn on Figure 7 at $d = 1.0 \times 10^7$ m ✓ Calculation of gradient of a straight line ✓ Their gradient divided by 37 kg to give 0.039. ✓	Only penalise POT in final answer Expect to see gradient = 1.4 and field strength = $0.039 \text{ (N kg}^{-1}\text{)}$ Allow 0.035 to 0.044 with working. If no other mark awarded allow MAX 1 for evidence of their E_p divided by 37 kg.	3	AO3 × 2 AO2 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Determines maximum GPE from Figure 7 ✓ Equates change in GPE (from Moon to max) to KE (at Moon) OR Uses KE equation with any value of E_p to determine their v ✓ Answer that rounds to $2.2 \times 10^3 \text{ m s}^{-1}$ ✓	Accept (–) $4.7 \times 10^7 \text{ (J)}$ to $4.9 \times 10^7 \text{ (J)}$ seen Expect 9.1 to $9.3 \times 10^7 \text{ (J)}$ for change in GPE Do not allow approach based on escape velocity of the Moon.	3	AO3 × 1 AO2 × 2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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11	D	$\frac{4\pi^2}{GM}$	AO1

19	B	$3.6 \times 10^7 \text{ m}$	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Attempt to use $F = G \frac{m_1 m_2}{r^2}$ ✓ 120 (N) ✓	Require to see at least mass of B and either r^2 for mp1. Condone POT errors. Correct answer gets both marks. Accept answer that rounds to 120 to 3 sf OR 120.5 NOTE answer using incorrect $r = 11.3$ (N) gets 1 mark	2	1 × AO3 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Any two from: ✓✓ • $E_p = (-)G \frac{m_1 m_2}{r}$ seen • evidence of $E_k = -1.19 \times 10^{12}$ – their E_p • attempts to use $\frac{1}{2} m v^2 =$ their E_k $2.1 \times 10^4 \text{ (m s}^{-1}\text{)} \checkmark$	Condone missing or incorrect minus signs in mp1 and mp2. Do not reward equations based on a circular orbit	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	gravitational force = mass $\times$ centripetal acceleration seen in some form OR Evidence of $\omega = \frac{2\pi}{T}$ OR $v = r\omega$ in an expression for centripetal acceleration ✓ ₁ $r = \sqrt[3]{\frac{GMT^2}{4\pi^2}} \quad \checkmark_1 \checkmark_2$ $8.9 \times 10^{10} \text{ (m)} \checkmark_3$	Award mp1 and mp2 for correct radius equation seen For MP2 accept $4\pi^2 r^3 = GMT^2$ in any form	3	1 $\times$ AO3 1 $\times$ AO1 1 $\times$ AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Idea that the (gravitational) potential (energy of the spacecraft) is constant on an equipotential surface so that B is not on an equipotential as its radius changes OR C is on an equipotential surface as its radius is constant.	Idea can be inferred Accept idea that in a circular orbit the gravitational force is at right angles to the motion of satellite so no work is done.	1	AO1

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	The work done per unit mass ✓ in moving a (small test) mass from infinity to the point ✓		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Any one from: ✓ • halve the potential or mass • use of $M = -\frac{V_r}{G}$ Mass of star A = 8.0×10^{30} (kg) ✓		2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Change in potential / change in distance $\checkmark_1$ Tangent drawn at their distance (range tbc) AND used for gradient with at least 4 squares on shortest side $\checkmark_2$ $g_T = 1.8 \times 10^{-4} \text{ (N kg}^{-1}\text{)}$ AND $x = 1.1 \times 10^{-12} \text{ (m)}$ $\checkmark_3$ (tolerance tbc)	[Show answer of 4.2] Condone misreads and data taken from curve for $\checkmark_1$ Condone misreads and missing or incorrect minus signs for gradient value in $\checkmark_2$ Alternative Read off max value from graph $\checkmark_1$ Use of trig to get component $\checkmark_2$ $g_T = 1.8 \times 10^{-4} \text{ (N kg}^{-1}\text{)}$ AND $x = 1.1 \times 10^{-12} \text{ (m)}$ $\checkmark_3$ (tolerance tbc)	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Gravitational field strength is 0 at O $\checkmark$ Increases in magnitude (to a maximum) as x increases, then decreases in magnitude as x increases further $\checkmark$		2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Any two from: ✓✓ • straight line drawn as tangent to line at $\ln x = 4.0$ or statement that line has to be straight or use of logged equation aligned with $y = mx + c$ or idea that gradient = -2 • read-off where their straight line and curve diverge or attempt at read-off where line stops being straight • use of e^x with their read-off $x_{\min} = 4.5\text{--}5.5 \times 10^{12} \text{ (m)}$ ✓		3	AO3
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
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10	A	$\frac{R}{4}$	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	The (gravitational) field from the Earth and the Moon are acting in opposite directions AND are equal in magnitude ✓	Allow (gravitational) force instead of field	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Equating $g = \frac{GM}{r^2}$ for Earth and Moon at P ✓ $7.2 \times 10^{22} \text{ (kg)} \checkmark$	$\frac{G \times 5.98 \times 10^{24}}{(3.47 \times 10^8)^2} = \frac{Gm}{(3.8 \times 10^7)^2}$ $m = 7.17 \times 10^{22} \text{ kg}$	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Energy is required to move (a mass) from P to infinity OR The gravitational potential due to the Earth and the Moon are both negative and so when added together will not cancel or add to zero (but give a larger magnitude total instead) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Evidence of tangent drawn on graph at $2.0 \times 10^8 \text{ m}$ ✓ Calculation of gradient shown ✓ Answer that rounds to 2 sf in range (-) 0.0085 to (-) 0.013 (N kg^{-1}) ✓	Allow POT error on mp2 Do not allow mp3 for answers in range but from an incorrect method shown	3	1 × AO1 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	$\Delta V = 6.26 \times 10^7 - 1.3 \times 10^6 (= 6.13 \times 10^7) (\text{Jkg}^{-1})$ ✓ $\Delta W = 950 \times \Delta V$ correctly calculated ✓	Allow in range 6.13 to 6.14×10^7 which relate to a read-off of 1.3×10^6 and 1.2×10^6 respectively For mp2, ecf can be given for the wrong change in potential, for example outside of above range, or for taking it between 6.26×10^7 and 4.0×10^6 , to the surface of the Moon on the graph (or 3.85×10^6 using data in question) Do not allow mp2 for a change in potential to zero (i.e. if $950 \times V$ is calculated) Expected answer = $5.8 \times 10^{10} \text{ J}$	2	1 × AO2 1 × AO3

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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10	B	The satellite accelerates towards the centre of the Earth.	AO1
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13	A	$T_X < T_Y$	$E_X < E_Y$	AU3
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Question	Marking guidance	Mark	Comments
01.1	$g = (-) \frac{GM}{r^2} \quad \checkmark$ $r (= 6.4 \times 10^6 + 2.04 \times 10^7) = 2.68 \times 10^7 \text{ (m)} \quad \checkmark$ $g = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{(2.68 \times 10^7)^2} \quad \checkmark \quad (= 0.56 \text{ N kg}^{-1})$	3	
01.2	$g = \frac{v^2}{r}$ $v = [0.56 \times (2.68 \times 10^7)]^{1/2} \quad \checkmark$ $= 3.9 \times 10^3 \text{ m s}^{-1} \quad \checkmark \quad (3.87 \times 10^3 \text{ m s}^{-1})$ (allow ecf for value of r from 01.1)	2	Allow alternative methods

Unknown Unknown - Unit 3 - Q01 (continued 2)

$$T \left(= \frac{2\pi r}{v} \right) = \frac{2\pi \times 2.00 \times 10^3}{3.87 \times 10^3} \quad \checkmark$$

$= 4.3(5) \times 10^4 \text{ (s)} \quad \checkmark \quad (12.(1) \text{ hours})$
 (use of $v = 3.9 \times 10^3$ gives $T = 4.3(1) \times 10^4 \text{ s} = 12.0 \text{ hours}$)
 (allow C.E. for value of v from 01.2)

Unknown Unknown - Unit 3 - Q01 (continued 3)

question	marking guidance	mark	comments
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